



HEXGUARD

A multi-source AI-powered network intrusion detection system

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STATUS	Active development
DATE	9/8/2026

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01 Executive summary

HexGuard is a layered network intrusion detection system combining signature-based detection (Suricata and YARA), supervised machine learning classification (XGBoost), unsupervised anomaly detection (Isolation Forest), and rule-based behavioral detectors mapped to the MITRE ATT&CK framework.

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DETECTION LAYERS — SIGNATURE, ML ×2, ANOMALY, BEHAVIORAL

~47M

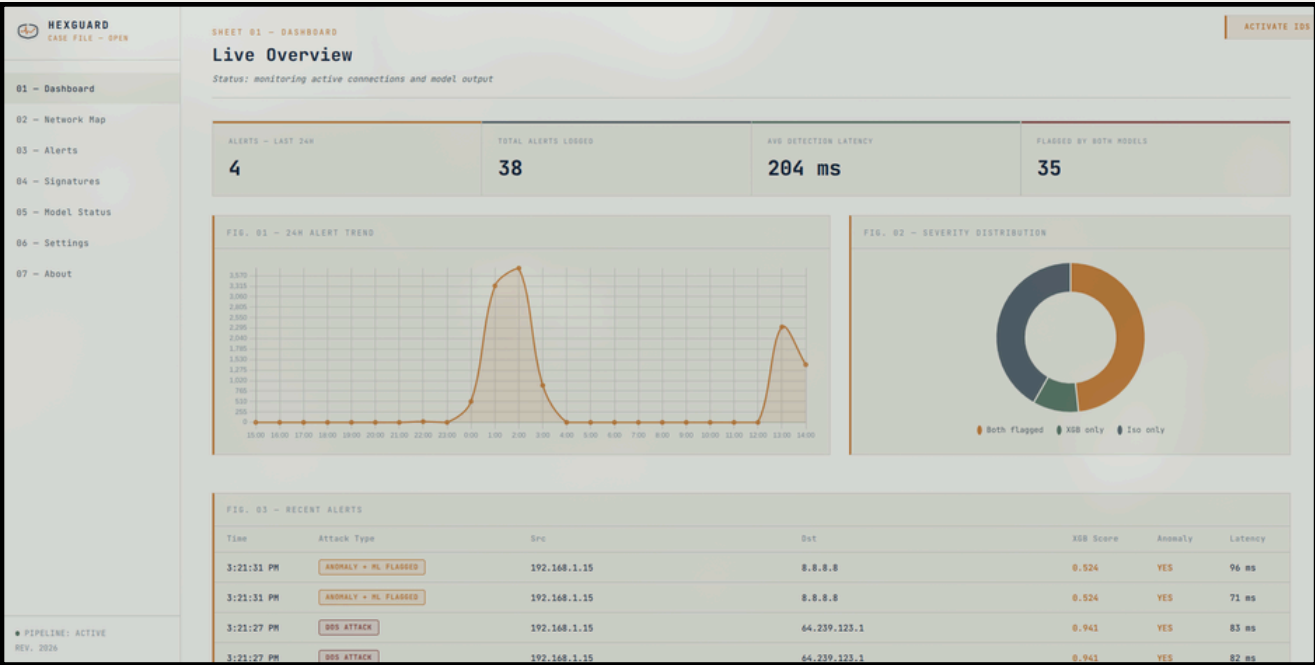
TRAINING ROWS ACROSS THREE MERGED BENCHMARK DATASETS

52,148

ETOPEN SIGNATURE RULES ACTIVE IN THE SURICATA LAYER

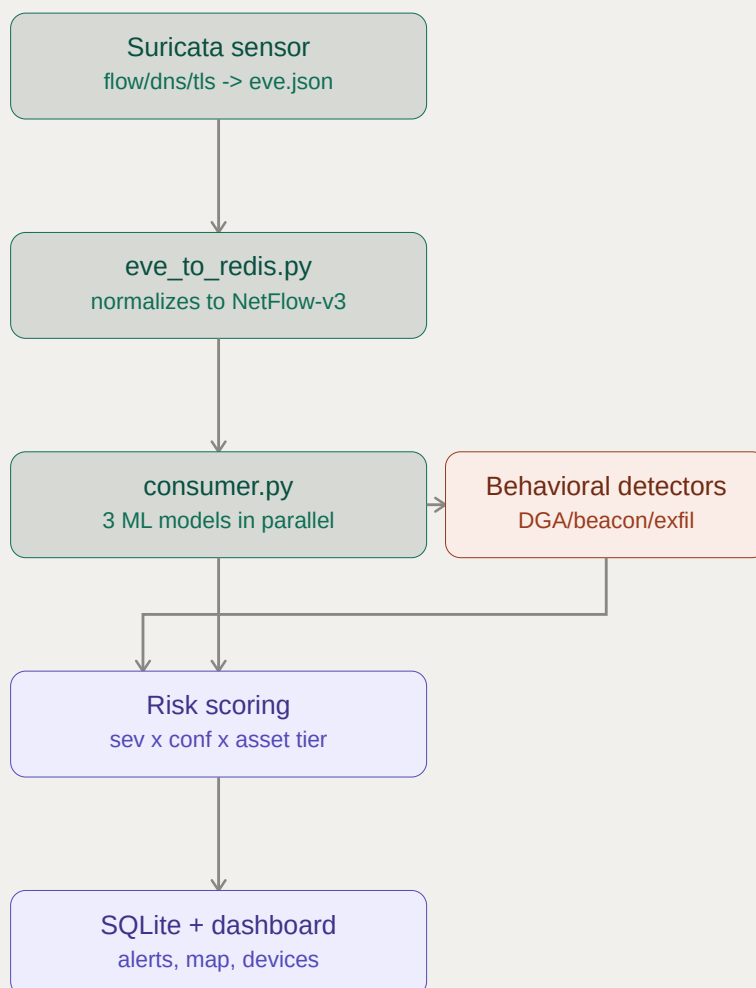
The most significant result produced by this project to date is not a detection accuracy figure. Cross-dataset validation revealed that the initial classification models were substantially learning testbed-specific artifacts — timestamp and topology signal — rather than universal attack behavior. This finding drove a feature-leakage investigation, a retrained model generation (v2), and an ongoing effort to close the gap between laboratory training conditions and live network capture. That investigation, and the discipline of not trusting in-dataset numbers, is presented here as the core engineering contribution of the project.

HexGuard is under active development. This document reflects the system's current state, including known limitations, and is updated as the retraining and hardening work progresses.



02 System architecture

HexGuard's pipeline is a linear, streaming path from raw traffic to an analyst-facing dashboard. Each stage is decoupled through Redis Streams, allowing the sensor, inference, and presentation layers to run, restart, and scale independently.



03 Datasets

Model training draws on three independent public benchmark datasets plus a small proprietary baseline capture, spanning both traditional enterprise network attacks and IoT / IIoT traffic.

Dataset	Size	Rows (approx.)	Classes	Role
CSE-CIC-IDS2018	~4.2 GB	~19.5 M	9	Primary training set
ToN-IoT	~5.3 GB	~25.7 M	10	IoT / IIoT traffic coverage
UNSW-NB15	~577 MB	~2.4 M	9	Cross-validation / generalization testing
Proprietary baseline	284 KB	697	BENIGN	Known-good anomaly reference

04 Detection layers

Signature-based layer

Suricata 6.0.20 runs the ET Open ruleset (52,148 rules) alongside custom local rules; YARA provides hash- and pattern-based malware detection. Catches known-bad traffic with high precision and low latency, and corroborates or escalates findings from the ML and behavioral layers.

SHEET 04 - SIGNATURES DEACTIVATE IDS

Suricata & YARA

Known-bad pattern matches – separate from the ML layers

FIG. 02 – SURICATA fast.log

500 alert(s) in fast.log (newest first).

TIME	PRIORITY	SID:SID-REV	MESSAGE	CLASSIFICATION	PROTO	SOURCE	DESTINATION
5:03:34 PM	P2	1:2100498:7	GPL ATTACK_RESPONSE id check returned root	Potentially Bad Traffic	TCP	217.160.0.187:80	192.168.1.18:47572
4:40:00 PM	P2	1:2100498:8	GPL ATTACK_RESPONSE id check returned root	Potentially Bad Traffic	TCP	217.160.0.187:80	192.168.1.15:43160
4:35:16 PM	P2	1:2100498:7	GPL ATTACK_RESPONSE id check returned root	Potentially Bad Traffic	TCP	217.160.0.187:80	192.168.1.15:54552
4:32:25 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:56111	74.125.250.129:19302
4:32:24 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:55571	74.125.250.129:19302
4:32:14 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:56111	74.125.250.129:19302
4:32:13 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:55571	74.125.250.129:19302
4:32:04 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:56111	74.125.250.129:19302
4:32:03 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:55571	74.125.250.129:19302
4:31:54 PM	P3	1:2033078:5	ET INFO Session Traversal Utilities for NAT (STUN Binding Request On Non-Standard High Port)	Misc activity	UDP	192.168.1.15:56111	74.125.250.129:19302

Binary multi-source classifier

An XGBoost model trained on a merged CSE-CIC-IDS2018 + ToN-IoT + UNSW-NB15 dataset (~47M rows combined), collapsing the decision to a binary ATTACK / BENIGN outcome. Pooled accuracy ≈ 0.907, ATTACK-class F1 ≈ 0.943 on the combined set.

SHEET 03 - ALERTS DEACTIVATE IDS

Case Log

Filter, review, and triage detections from both models

src ip contains... dst ip contains... min xgb score ☐ hide false positives APPLY

9056 NETWORK ANOMALY 569 PORT SCAN 445 INFORMATION EVENT 411 ANOMALY (UNCLASSIFIED) 41 INFILTRATION 38 BEACONING_C2 10 DDoS ATTACK

CONFIRMED ATTACKS

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5:09:27 PM	NETWORK ANOMALY	192.168.1.18 - 35.223.238.178 (TCP)
5:09:23 PM	NETWORK ANOMALY	fe80:0000:0000:0000:f0ab:00ff:febe:ee9a - ff02:0000:0000:0000:0000:0000:0000:00fb (UDP)
5:09:19 PM	NETWORK ANOMALY	192.168.1.11 - 224.0.0.251 (UDP)
5:09:03 PM	NETWORK ANOMALY	fe80:0000:0000:0000:f0ab:00ff:febe:ee9a - ff02:0000:0000:0000:0000:0000:0000:00fb (UDP)
5:09:03 PM	NETWORK ANOMALY	192.168.1.11 - 224.0.0.251 (UDP)
5:08:54 PM	NETWORK ANOMALY	192.168.1.1 - 239.255.255.250 (UDP)
5:08:49 PM	NETWORK ANOMALY	192.168.1.18 - 140.82.112.26 (TCP)
5:08:43 PM	NETWORK ANOMALY	fe80:0000:0000:0000:f0ab:00ff:febe:ee9a - ff02:0000:0000:0000:0000:0000:0000:00fb (UDP)
5:08:40 PM	NETWORK ANOMALY	192.168.1.18 - 8.8.8.8 (UDP)
5:08:39 PM	NETWORK ANOMALY	192.168.1.11 - 224.0.0.251 (UDP)

Anomaly detection layer

An Isolation Forest trained exclusively on benign traffic provides an unsupervised "unclassified / zero-day" layer for traffic that matches no known signature or trained class — flagging genuinely novel behavior rather than only what it was explicitly trained to recognize.

198556	5:03:18 PM	ANOMALY + ML FLAGGED	192.168.1.18	172.67.69.147	TCP	0.992	YES	174 ms	no
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Rule-based behavioral detectors

Four heuristics run independently of the ML layer, each mapped to a MITRE ATT&CK technique, targeting post-compromise behavior classification models are not well suited to catch:

- **DGA detection** T1568.002 — NXDOMAIN rate $\geq$ 40% (min. 20-query sample), entropy $\geq$ 3.5 bits, query length $>$ 20 chars, $\geq$ 5 high-entropy domains in a 600s window. Fires on lookup pattern, not a domain blocklist — survives DGA algorithm changes.
- **Beaconing / C2** T1071 / T1573 — periodic, fixed-size callback traffic via JA3 TLS fingerprinting against known C2 stacks.
- **Exfiltration** T1041 / T1048 — abnormal outbound volume and protocol misuse.
- **Lateral movement** T1021.x — SMB / RDP / SSH spread patterns across multiple hosts.

POST-COMPROMISE DETECTORS (PHASE 6)		50
Beaconing / C2 T1071		
192.168.1.18 → 140.82.112.26		
periodic beaconing suspected (CV=0.078, 23 conns)		
5:24:13 PM	LOW	
Beaconing / C2 T1071		
192.168.1.18 → 140.82.112.26		
periodic beaconing suspected (CV=0.078, 23 conns)		
5:22:57 PM	LOW	
Beaconing / C2 T1071		
192.168.1.18 → 140.82.112.26		
periodic beaconing suspected (CV=0.078, 22 conns)		
5:21:40 PM	LOW	
Beaconing / C2 T1071		
192.168.1.18 → 140.82.112.26		
periodic beaconing suspected (CV=0.078, 21 conns)		
5:20:23 PM	LOW	

05 The generalization finding

This section documents the project's central technical finding: strong in-dataset performance did not hold when the models were evaluated against independent data sources — and the reason why was traced to a specific, fixable cause.

In-dataset results looked strong

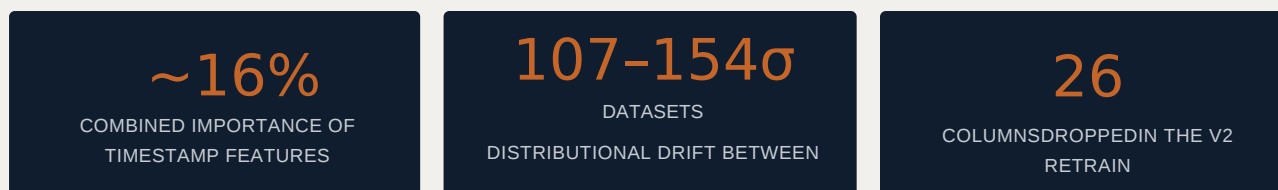
Trained and evaluated within a single dataset, both the multiclass and binary classifiers produced results in line with published benchmark expectations (Section 5).

Cross-dataset validation exposed the collapse

Evaluation set	Accuracy	ATTACK F1
ToN-IoT	0.5217	0.1187
UNSW-NB15	0.7543	0.0304

Evaluated against traffic sources the model had not been trained on, accuracy and attack-class F1 both dropped sharply — a strong signal that the model had learned something other than "is this traffic malicious."

Root cause: timestamp and topology leakage



Feature importance and SHAP analysis identified that timestamp-derived features (FLOW_START_MILLISECONDS / FLOW_END_MILLISECONDS) carried the largest single feature-importance share, with severe distributional drift between datasets. In effect, the model was substantially learning *which lab environment generated a given flow* rather than whether that flow was malicious.

Mitigation and current status

A second model generation (v2) was retrained with absolute timestamps and 26 additional non-live-derivable columns removed, specifically to reduce leakage and improve deployability against real Suricata-captured traffic. This validation work directly informs an ongoing retraining effort aimed at closing the remaining gap between laboratory-trained features and what is actually observable in live network capture.

06 Live pipeline & dashboard

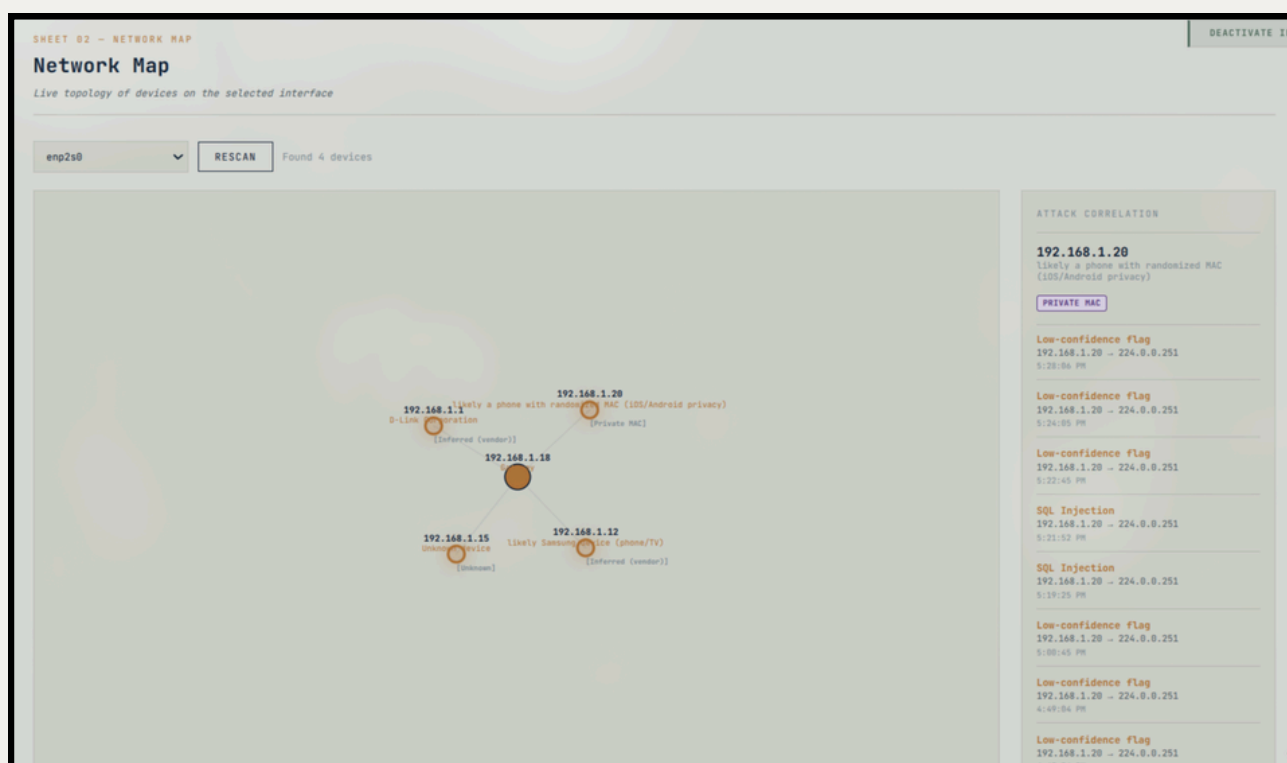
HexGuard runs as a real-time pipeline from Suricata capture to an analyst-facing desktop dashboard built on Electron. Alerts stream through Redis to a SQLite-backed store, and the dashboard surfaces them across dedicated panels for alerts, network topology mapping, device inventory, and pipeline control.

Risk scoring

severity × confidence × asset-criticality tier reorders alerts by operational priority rather than raw arrival order.

Deduplication

A 300-second window collapses repeated events of the same type between the same hosts into a single entry.



07 Limitations & future work

- **Cross-dataset generalization gap** — in progress; being addressed through continued feature-leakage elimination and retraining, per Section 6.
- **Train / serve feature mismatch** — the feature set available from live Suricata capture differs from the full offline dataset feature set. A feature-shim layer bridges this at inference time; models are being retrained specifically on the reduced, live-derivable feature set to reduce false positives on production-like traffic.
- **Planned work** — expanded behavioral detector coverage, tighter integration between analyst false-positive labeling and model suppression logic, and continued cross-dataset benchmarking as new data sources are added.